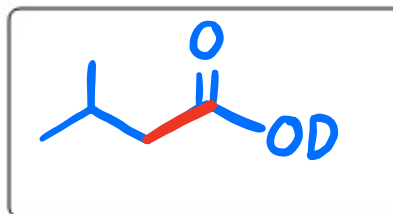
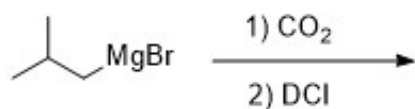


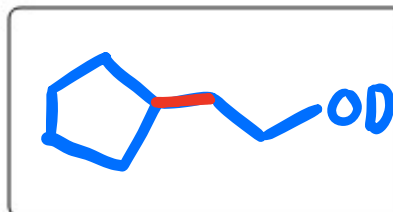
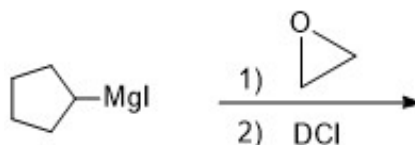
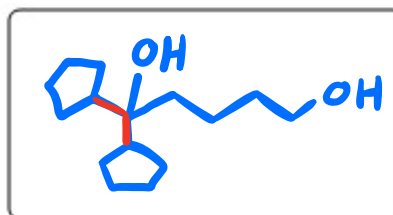
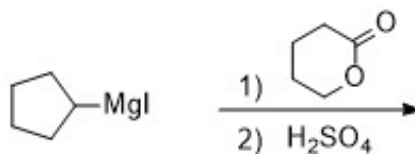
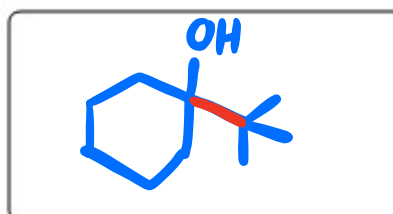
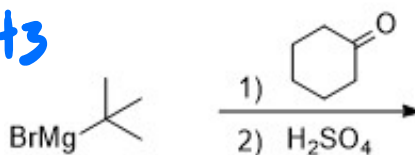
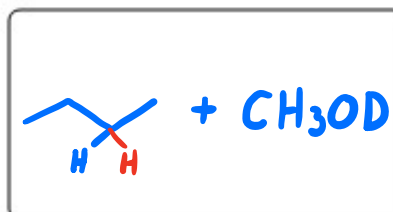
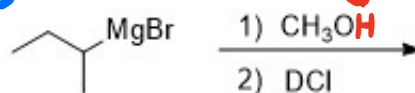
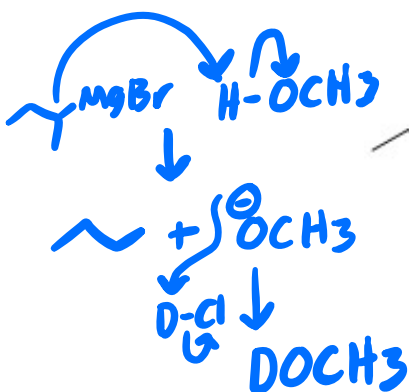
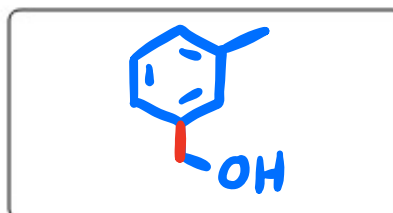
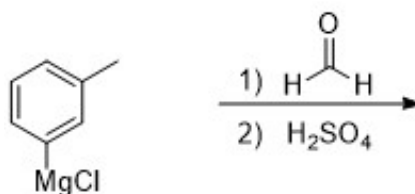
Post-midterm review material

Activity 7: Grignard Reactions

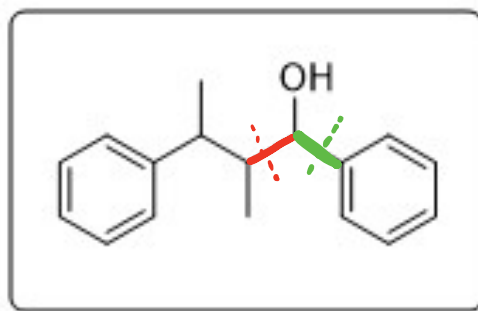
1. Predict the product of the following reactions.



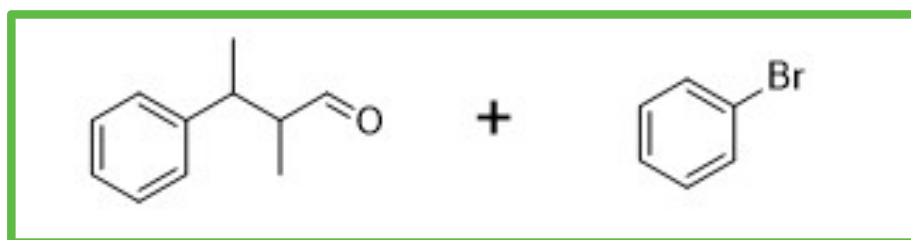
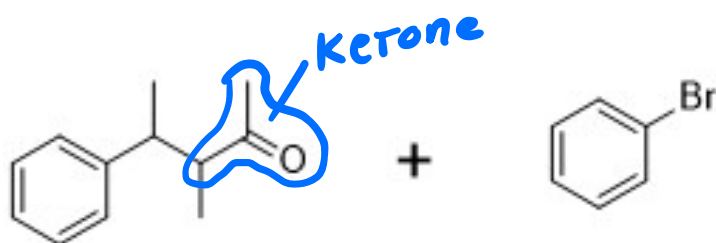
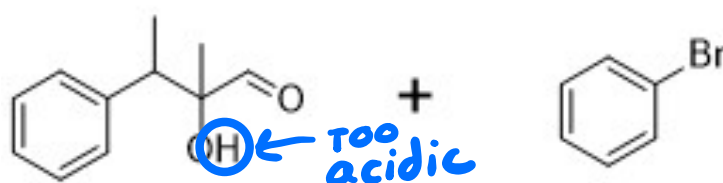
Bond that is red is a new C-C bond in each



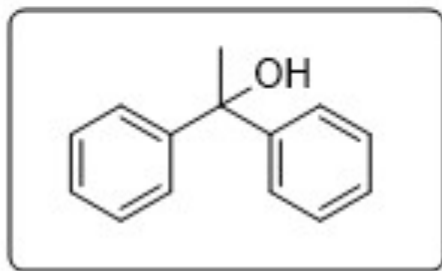
2. Which of the following combination of reactants could be used to yield the product in the box? Assume acidic workup.



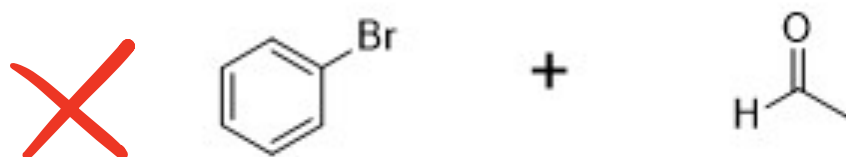
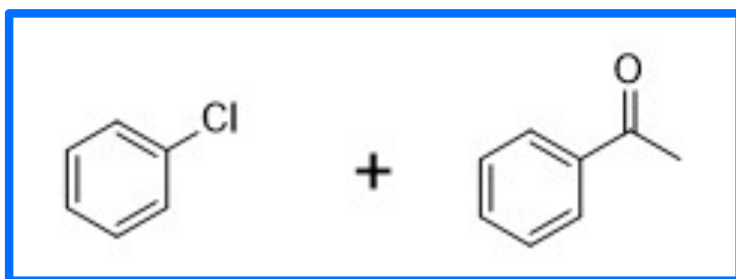
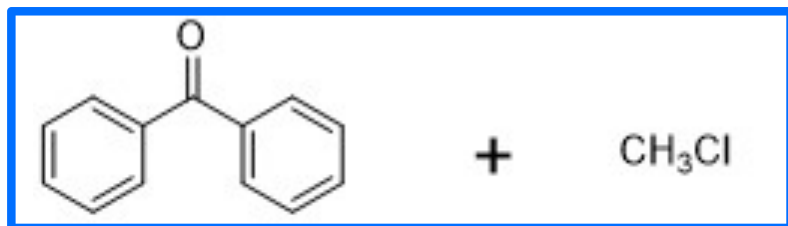
★ 2° alcohol. Look for an organohalide to form a Grignard and the electrophile should be an aldehyde but not formaldehyde



★ 3° alcohol. Look for ketone electrophiles

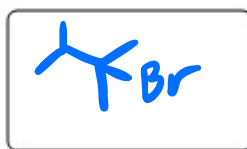
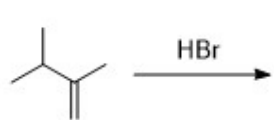


★ 3° alcohol with two equal R groups, look for correct ester electrophile

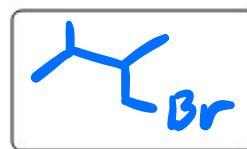


Activity 8: Alkene Addition Reactions

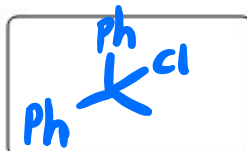
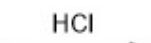
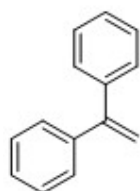
3. Predict the Markovnikov and anti-Markovnikov products of each of the following alkene addition reactions.



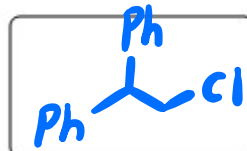
Markovnikov



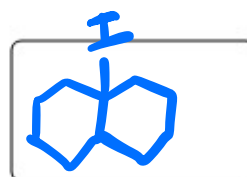
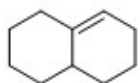
anti-Markovnikov



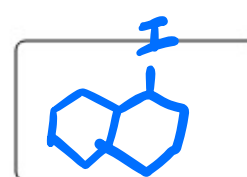
Markovnikov



anti-Markovnikov

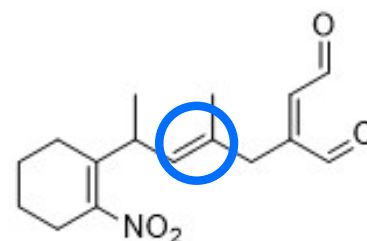
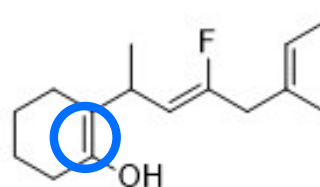
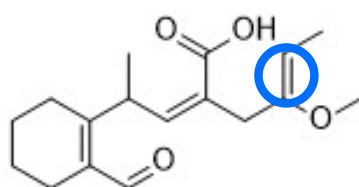
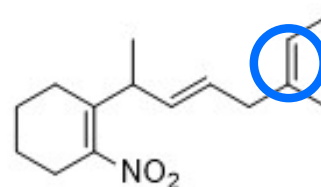
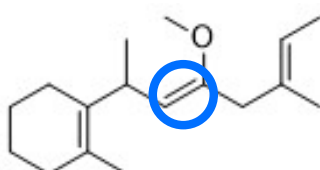
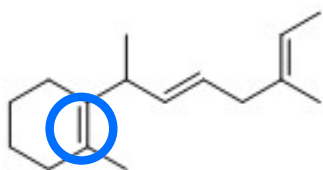


Markovnikov

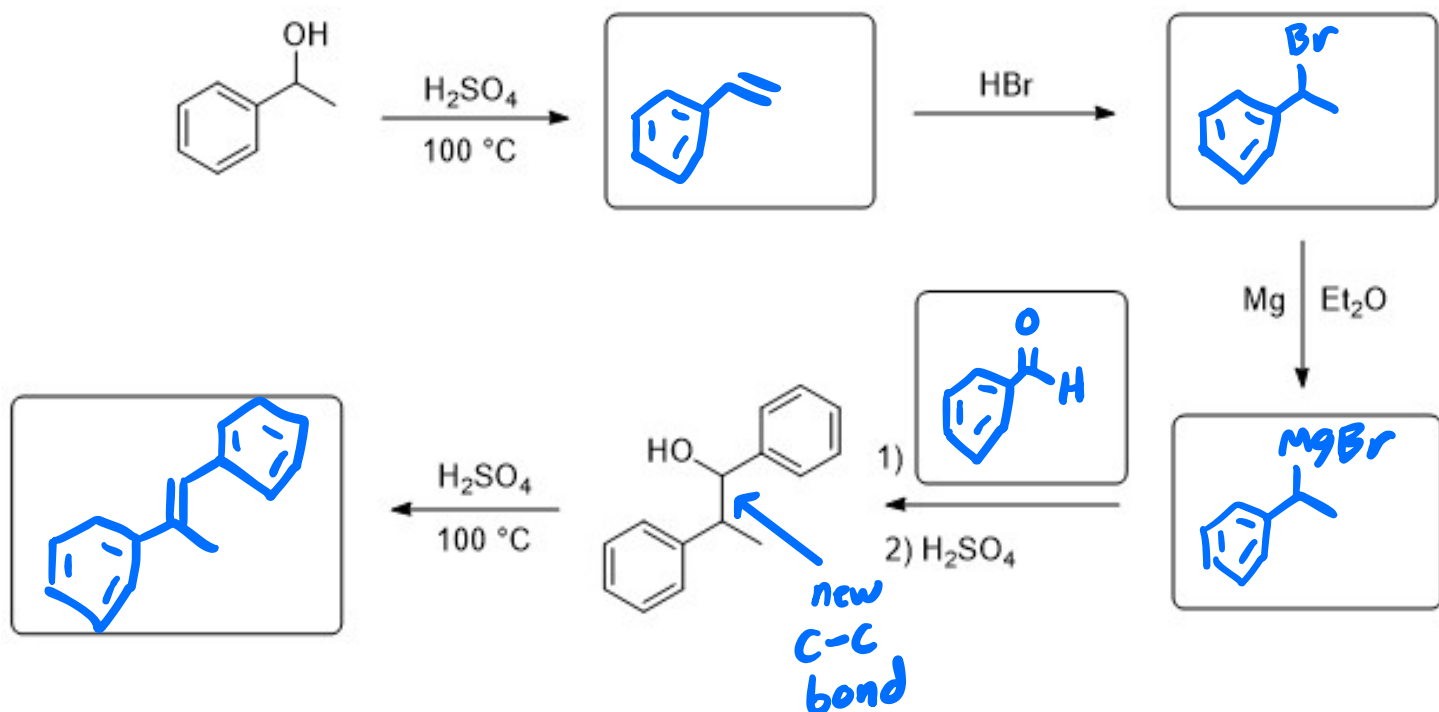


anti-Markovnikov

4. In each of the following triene molecules, circle the double bond that is most reactive towards addition of a hydrogen halide.

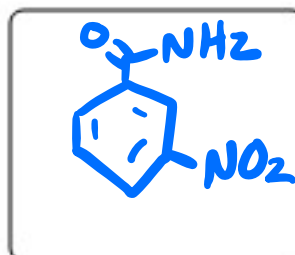
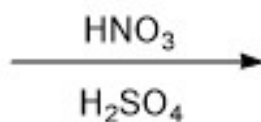
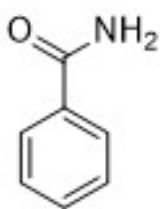
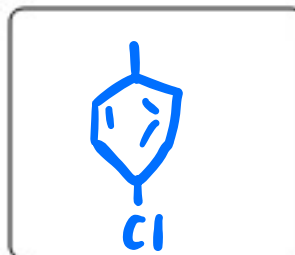
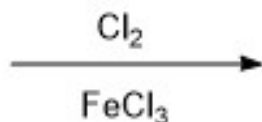
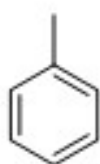


5. Fill in the boxes with the missing reagent/product.

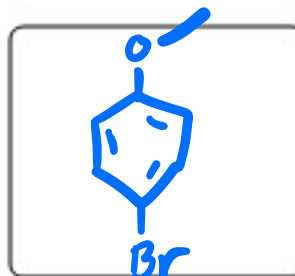
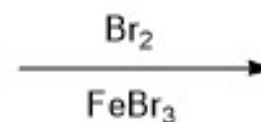
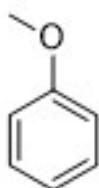
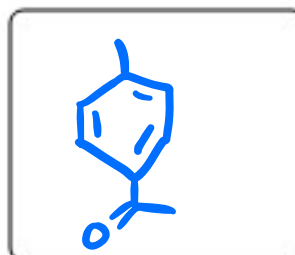
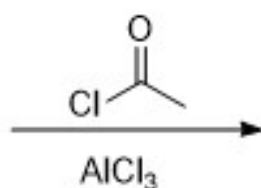
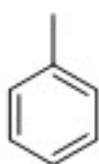
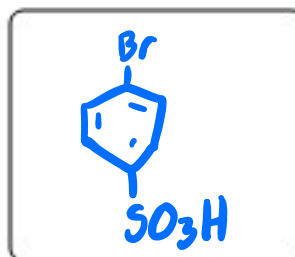
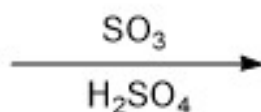
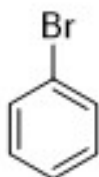


Activity 9: EAS Reactions

6. Predict the **MAJOR** product of each of the following reactions.

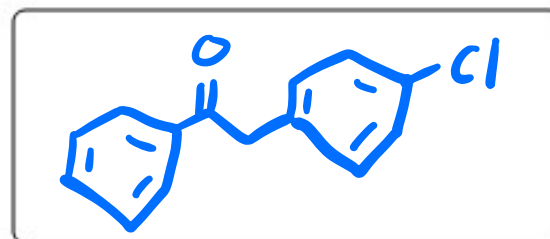
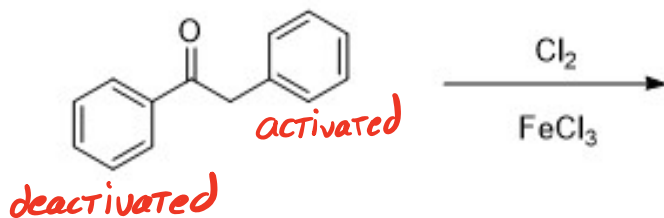
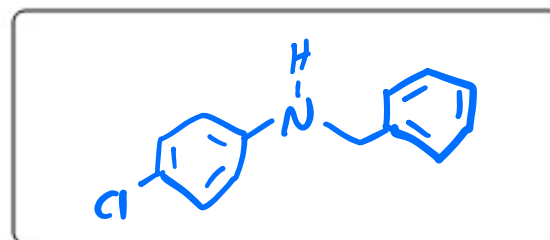
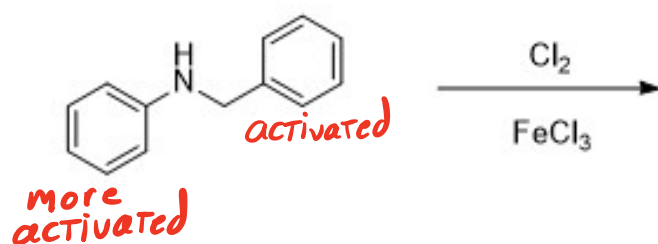
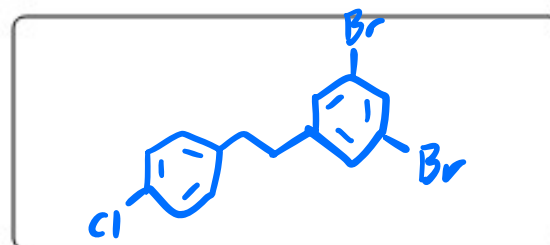
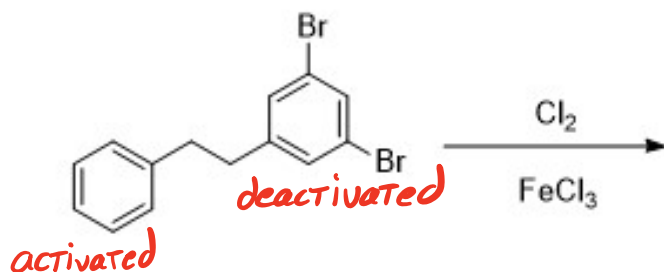
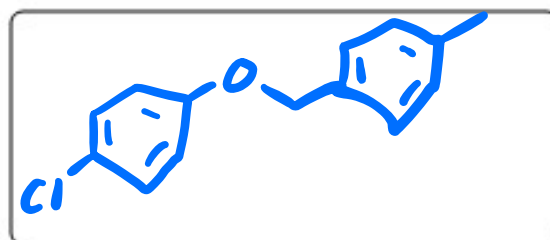
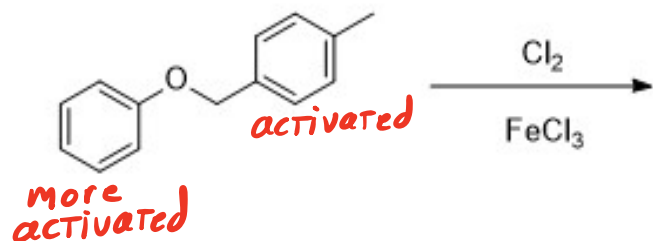
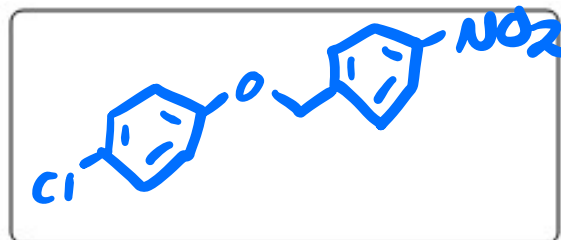
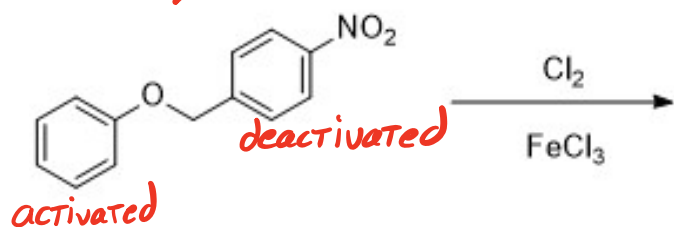


★
Halogens are
deactivators
but ortho/para
directors



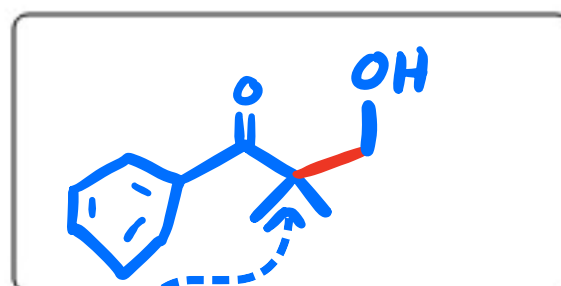
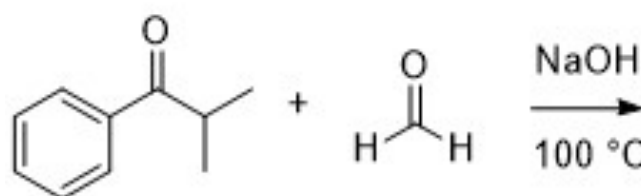
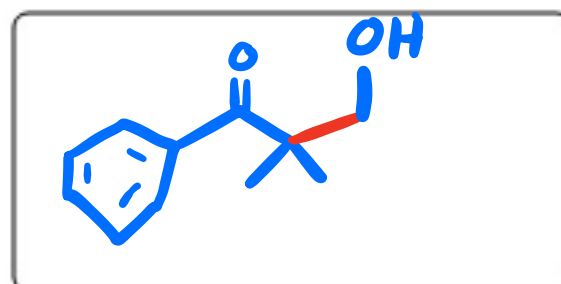
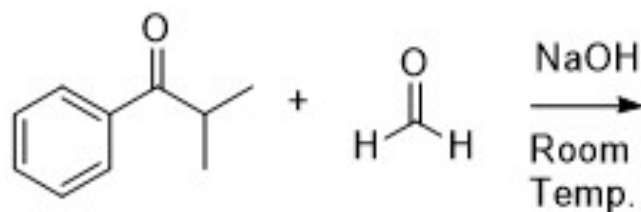
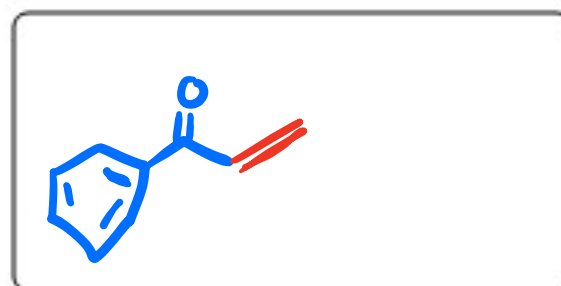
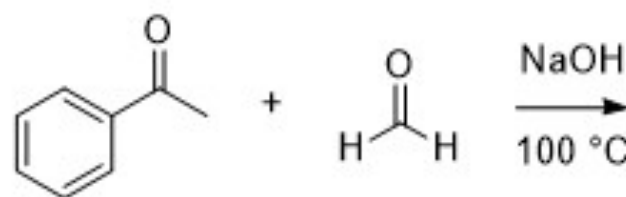
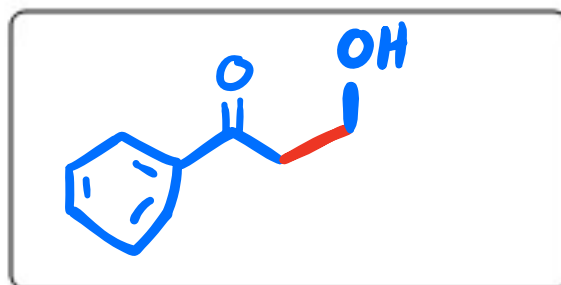
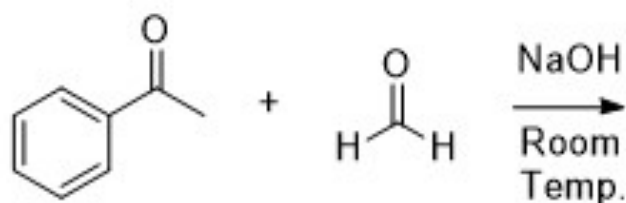
7. Predict the **MAJOR** product of each reaction chlorination reaction. Assume monochlorination.

* Reaction will occur at most activated ring



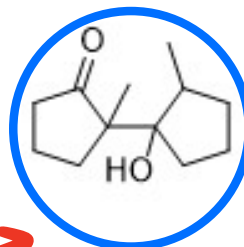
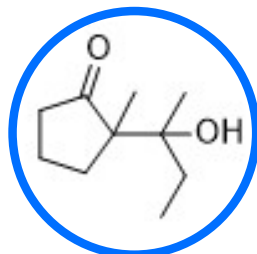
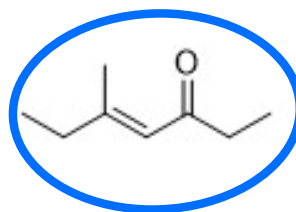
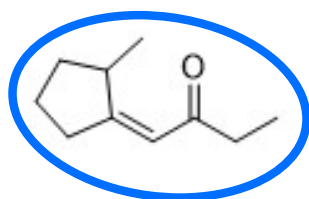
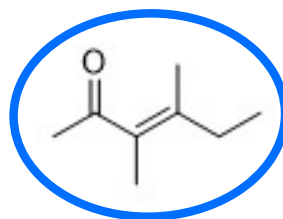
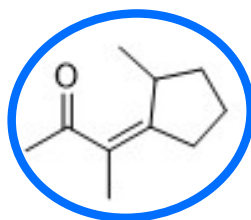
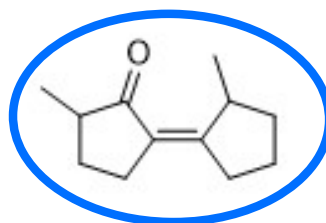
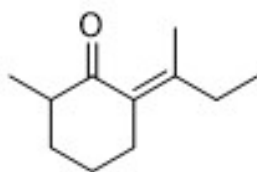
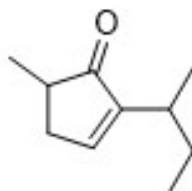
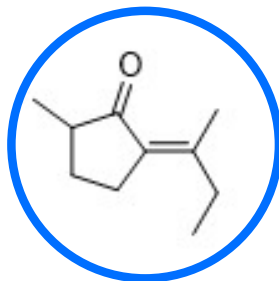
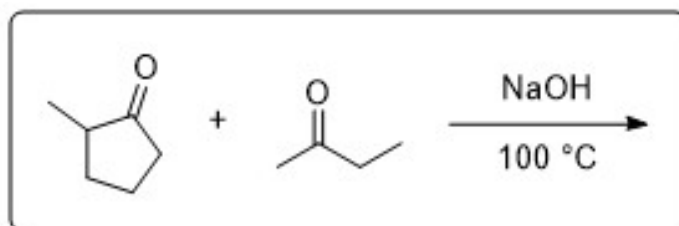
Activity 10: Aldol Addition & Condensation Reactions

8. Predict the **MAJOR** product of each of the following reactions. ** new C-C bond shown in red*



no H left. Can't form condensation product

9. Circle **all possible** product that can be formed from the following aldol condensation reaction.



*These two aldols are possible
but can't form the condensation product*